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Good and Bad Examples Of Engineering Processes

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Outline

- Introduction
- Bad examples of Engineering Processes
- Good examples of Engineering Processes
- Conclusion

Introduction

- Engineering processes help ensure quality, safety and efficiency.
- Well-designed processes reduce errors and improve communication.
- I will show real examples of bad and good engineering processes.

Tacoma Narrows Bridge Collapse, USA 1940



- The Tacoma Narrows Bridge opened in 1940 and immediately showed unusual movement even in normal wind.
- On November 7th, strong winds caused the bridge to twist violently from side to side.
- After several minutes of extreme motion, the bridge broke apart and collapsed into the river.

Tacoma Narrows Bridge Collapse, USA 1940



- Engineers did not test how the wind would affect the bridge's structure.
- The bridge was designed too flexible, which made it move excessively during strong winds.
- It became unstable and eventually broke apart.

Space Shuttle Challenger Explosion 1986



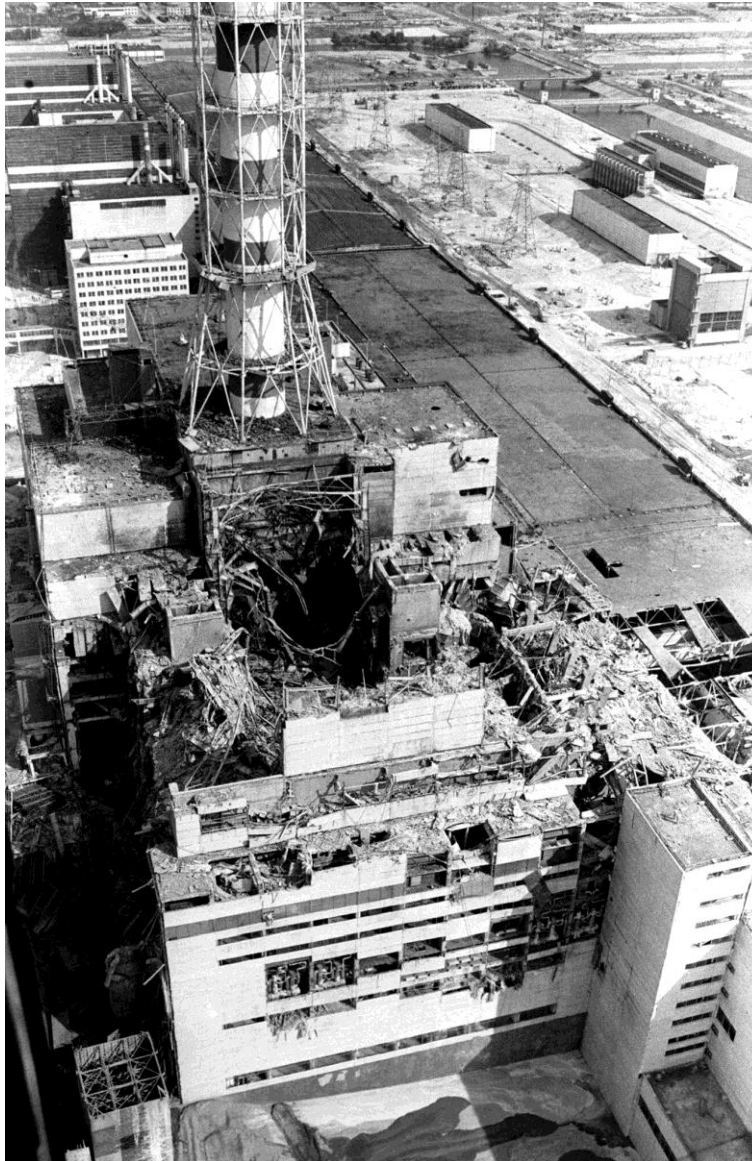
- The Challenger Space Shuttle launched on a very cold morning in January 1986.
- Seventy-three seconds after take-off, the shuttle broke apart in the air.
- All seven astronauts on board died in the accident.

Space Shuttle Challenger Explosion 1986



- The O-ring seal became too rigid in the cold and did not close the rocket joint properly.
- This failure allowed extremely hot gas to escape from the rocket booster.
- The gas leak caused a fire that led to the shuttle exploding during launch.

Chernobyl Nuclear Disaster 1986



- A safety test was performed at night in Reactor 4 of the Chernobyl nuclear plant.
- During the test, the reactor became unstable and its power increased uncontrollably.
- The reactor exploded and released radioactive material into the atmosphere.

Chernobyl Nuclear Disaster 1986



- Operators turned off important safety systems during the test.
- The reactor design was unstable at low power and reacted dangerously.
- Poor decisions and lack of training caused the explosion.

Deepwater Horizon Oil Spill, 2010



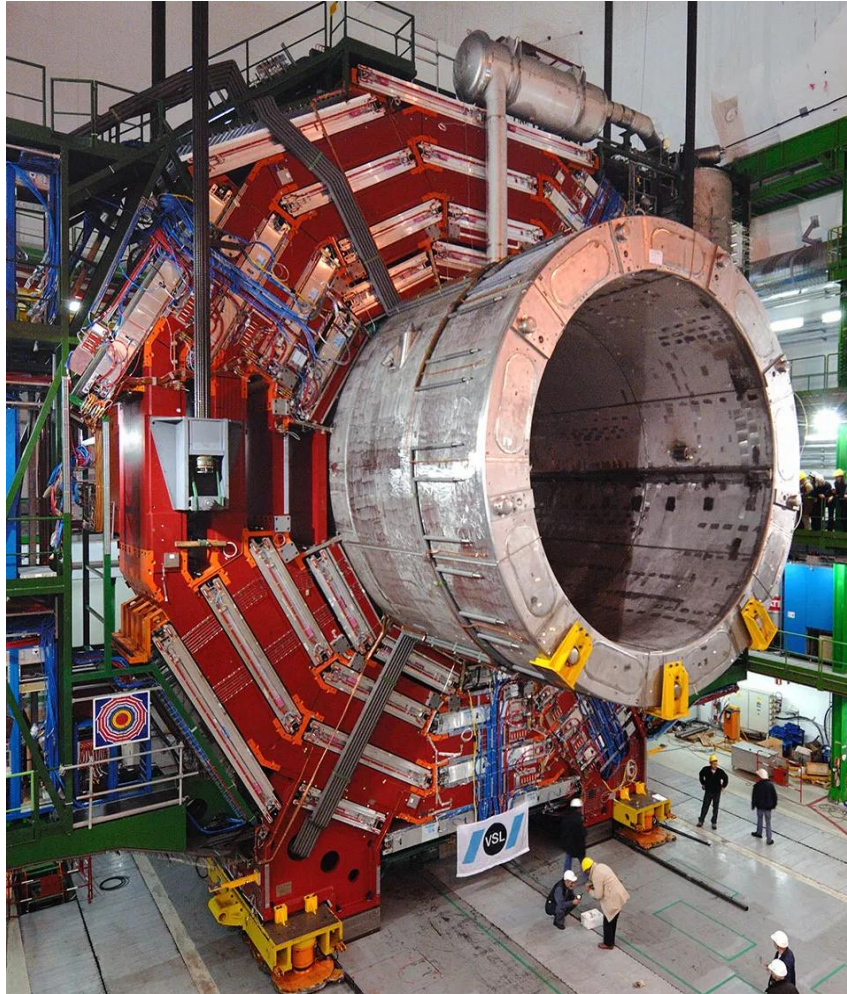
- The Deepwater Horizon was an offshore oil platform in the Gulf of Mexico.
- In April 2010, a sudden explosion occurred, followed by a massive fire.
- The platform sank, causing the largest marine oil spill in history.

Deepwater Horizon Oil Spill, 2010



- A safety device called the blowout preventer failed and did not stop the high-pressure oil.
- Warning signs from pressure tests were ignored before the explosion.
- Safety procedures were not followed, which allowed the disaster to happen.

Large Hadron Collider



source: Britannica

- The Large Hadron Collider is the world's largest particle accelerator, built in Switzerland.
- It is a 27-kilometre tunnel where particles are accelerated close to the speed of light.
- The machine allows scientists to study the smallest particles in the universe.

Large Hadron Collider



- Engineers built the tunnel and magnets with extreme precision so everything stays perfectly aligned.
- The system was tested carefully to work at extremely low temperatures near absolute zero.
- The project succeeded because the engineering process followed strict planning, testing and verification.

source: Wired

International Space Station



source: wikipedia

- The International Space Station was built by 15 countries over several years.
- Each module was launched separately and then connected in space.
- Astronauts live and work inside the ISS for long periods.

International Space Station



source: the week

- All modules were designed with strict standards so they could fit together perfectly.
- The ISS stays safe thanks to continuous monitoring and maintenance.
- It succeeded because the engineering process required strong cooperation and detailed planning.

Conclusions

- These examples show that when engineering processes fail, the consequences can be serious and sometimes catastrophic.
- All the accidents happened because important steps like planning, testing, communication or safety procedures were not followed correctly.
- Understanding these failures helps engineers improve their work and avoid repeating the same mistakes in future projects.



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Thank you!